Gadi Thanmayee

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Objective:

Motivated AI/ML Developer and Undergraduate engineering student in Computer Science & Engineering (AI & ML) at MLRITM, Hyderabad. Secretary of the Robotics & Automation Lab (RAL Club). Hands-on experience across Computer Vision, Deep Learning, Generative AI, RAG pipelines, and intelligent workflow automation. Currently working as an AI/ML Engineer at SustiVision AI, completed a three-month AI/ML internship at Qugate Technologies, and serving as ML Developer (Part-Time) at Oorjith. Passionate about turning trained machine learning models into robust, production-ready software systems and real-world architectures.

Education:

Marri Laxman Reddy Institute of Technology and Management [MLRITM], Sept 2024 – Jun 2028
Btech - Computer Science and Engineering [AIML]

Intermediate, Narayana Junior College, Kukatpally July 2024

Skills :

Programming: Python, C, C++, Java

Machine Learning: Scikit-learn, Pandas, NumPy, Model Evaluation, Feature Engineering, Supervised Learning

Deep Learning: PyTorch, TensorFlow, Keras, CNNs, Transfer Learning, Neural Networks

Computer Vision: OpenCV, YOLOv8, Object Detection, Image Classification, Foliar Diagnostics

Generative AI & Agentic AI: LLMs, RAG Pipelines, Prompt Engineering, AI Agents, Tool Calling, NLP, Vector Embeddings

Frameworks & APIs: FastAPI, Flask, Streamlit, Google Gemini API, REST APIs

Tools & Platforms: Git & GitHub, VS Code, Antigravity, Linux, PyAutoGUI

Databases: SQL, SQLite, Vector Databases (MongoDB Vector Search)

Professional & Interpersonal: Problem-solving, Implementation Planning, Technical Presentation, Team Collaboration

Experience:

AI/ML Engineer (Part-Time)

SustiVision AI | Nigeria, Africa • Present

- Serving as an AI/ML Engineer, applying and advancing practical expertise across artificial intelligence, machine learning, and computer vision systems.
- Designing and executing practical implementations of AI/ML architectures, exploring alternative algorithmic approaches and innovative problem-solving methods.
- Contributing actively to project development, model experimentation, and continuous technical improvement with a focus on meaningful real-world outcomes.
- Developing intelligent software solutions across modern AI/ML frameworks, tools, and scalable inference environments.
- Approaching complex engineering challenges from multiple perspectives to construct and refine robust AI-driven applications.

AI/ML Intern

Qugate Technologies | Bengaluru, Karnataka • June 2026 – September 2026 (Completed)

- Completed a three-month AI/ML internship gaining practical experience in AI/ML implementation, debugging, and productionizing models.
- Worked on the AI/ML platform for LEGOPHONICS, a hydroponics-based indoor farming system, collaborating closely with hardware engineers.
- Developed deep learning and computer vision models for crop disease detection and plant health diagnostics to enhance agronomic decision-making.
- Engineered intelligent image classification models and software automation solutions to streamline operational workflows and reduce manual tasks.
- Performed dataset preprocessing, model training, validation, performance benchmarking, and system integration.
- Cultivated hands-on expertise in the end-to-end AI development lifecycle, from data curation to testing and deployment.

ML Developer (Part-Time)

Oorjith | Chandigarh, Punjab (Remote) • July 2026 – Present

- Develop and support machine learning solutions by applying data preprocessing, exploratory analysis, and feature engineering techniques.
- Collaborate with cross-functional engineering teams to build, optimize, and evaluate AI-driven applications using Python and machine learning frameworks.
- Assist in analysing datasets, tuning model hyperparameters, and delivering practical AI solutions for production use cases.
- Gain hands-on experience across the entire machine learning workflow, from feature engineering to model deployment and runtime optimization.

Secretary — Robotics & Automation Lab (RAL Club)

MLRITM, Hyderabad • 2024 – Present

- Directing club technical initiatives, workshops, innovation competitions, and bootcamps for 200+ engineering students.
- Coordinating student engineering teams for national hackathons, embedded systems prototyping, and autonomous drone demonstrations.
- Leading hands-on peer mentoring sessions in microcontrollers, sensor-actuator interfacing, and integrated software-hardware automation pipelines.

Projects:

AI Plant Health & Disease Detection (Patent Filed — Team Lead)

- Developed a computer vision and agentic diagnostic system classifying plant leaf diseases using deep CNNs and Zero-Shot Learning.
- Delivered automated agronomy treatment plans, continuous foliar monitoring, and adaptive agronomic decision support.
- Filed official patent with the Indian Patent Office (Application No: 202641109078) as Team Lead for the core diagnostic architecture.
- Engineered high-throughput inference endpoints using FastAPI microservices and PyTorch deep learning models.
- Tech Stack: Python, FastAPI, PyTorch, Computer Vision, Deep Learning, Agentic AI, Zero-Shot Learning, OpenCV.

Ultron — AI Desktop Automation Agent

- Developed an autonomous desktop AI assistant translating spoken and text instructions into real-time operating system actions.
- Implemented multi-step computer workflow automation, application control, and speech recognition pipelines.
- Engineered voice-controlled agentic loops using Speech AI, PyAutoGUI, tool calling, and structured task execution.
- Tech Stack: Python, Agentic AI, Voice AI, PyAutoGUI, Tool Calling, Speech Recognition, OS Automation.

Accident Detection & Road Safety Vision

- Built an AI-powered computer vision system detecting vehicular road accidents from surveillance video imagery in real time.
- Trained custom CNN and YOLOv8 models for accurate object detection and incident classification under diverse lighting conditions.
- Integrated automated emergency dispatch sending Telegram alerts with precise GPS coordinates to response teams.
- Deployed the detection system as an interactive Flask-based web application with real-time video feed processing.
- Tech Stack: Python, TensorFlow, Keras, YOLOv8, OpenCV, Flask, Telegram API.

Smart Irrigation & Rain Alert System (Winner — College Project Expo)

- Designed and implemented an automated agricultural hardware system monitoring soil moisture, temperature, humidity, and rainfall.
- Programmed Arduino and ESP8266 microcontrollers to control relay pumps and trigger an audio rain buzzer alert.
- Demonstrated the complete sensory pipeline and water conservation impact, winning 1st Place at the College Project Expo.
- Tech Stack: Arduino, ESP8266, IoT Sensors, Relay Modules, Embedded C/C++, Hardware Automation.

IoT Hardware Chatbot (RAG System)

- Developed a domain-specific Retrieval-Augmented Generation (RAG) assistant for IoT sensors, boards, and microcontrollers.
- Indexed technical datasheets into vector representations with Wikipedia API fallback for comprehensive knowledge retrieval.
- Leveraged LLMs to synthesize accurate, context-aware technical responses showcased through an interactive Flask web UI.
- Tech Stack: Python, RAG Pipelines, LLMs, Flask, NLP, Vector Databases, Wikipedia API.

YTBT Cardio NLP — ECG to Medical Report Translation

- Built a clinical multimodal deep learning architecture translating 12-lead ECG waveform images into structured medical reports.
- Combined a PyTorch 2D CNN feature extractor with a Transformer language model for sequential diagnostic report generation.
- Engineered a secure multi-role Flask web portal (Doctor, Patient, Technician, Admin) equipped with SQLite audit logs.
- Tech Stack: Python, PyTorch, Transformers, 2D CNN, NLP, Flask, SQLite.

StudyMind AI — Adaptive Learning & Mastery Navigator

- Developed an adaptive personalized study navigator generating tailored learning roadmaps and dynamic study timetables.
- Integrated Google Gemini API for automated curriculum generation and intelligent study guidance.
- Implemented a Random Forest ML engine to diagnose student concept mastery levels and dynamically inject remedial topics.
- Built a secure proctored anti-cheat exam portal with synchronized peer timers using Flask and SQLite.
- Tech Stack: Python, Flask, SQLite, Google Gemini API, Scikit-learn, Random Forest.

AI Workforce Productivity & Task Management System

- Built an intelligent productivity and task tracking platform featuring multi-role dashboards for Managers, Employees, and Admins.
- Integrated machine learning algorithms for work log anomaly detection, identifying irregularities and workflow bottlenecks.
- Implemented project analytics, progress visualization, and performance reporting using Flask and SQLite.
- Tech Stack: Python, Flask, SQLite, Machine Learning, Scikit-learn, Data Analytics.

Spam Email Detection Using Machine Learning

- Developed a supervised machine learning classification pipeline to accurately detect and filter spam emails.
- Conducted comprehensive text preprocessing, TF-IDF vectorization, and engineered 13 custom feature extractions.
- Connected the trained model to a live email account for automated real-time incoming message classification.
- Tech Stack: Python, Scikit-learn, Pandas, NumPy, Feature Engineering, Supervised Learning.

Patents & Intellectual Property:

UNCERTAINTY-AWARE MULTIMODAL CROP HEALTH DETECTION, FORECASTING AND ADAPTIVE MANAGEMENT SYSTEM

- Role: Team Lead
- Application No: 202641109078
- Filing Authority: Indian Patent Office (IP India)
- Filing Date: September 11, 2026
- Docket & CBR: Docket 143606 | CBR 74259
- Institution: Dept. of CSM, MLRITM, Hyderabad
- System Overview: Led the research and engineering team in architecting an uncertainty-aware multimodal diagnostic system. The system integrates high-resolution foliar computer vision with multi-sensor environmental inputs to detect crop diseases under ambiguous field conditions, forecast pathogen progression, and generate adaptive agronomy interventions.

Courses & Certifications:

- Building RAG Apps Using MongoDB — MongoDB | Retrieval-Augmented Generation & Vector Search — Completed
- Building AI-Powered Search with MongoDB — MongoDB | Vector Indexes & Semantic Querying — Completed
- Vector Search Fundamentals — MongoDB | High-Dimensional Embeddings & Similarity Search — Completed
- AWS Educate — Introduction to Generative AI — AWS Educate | Foundation Models, LLMs & Prompt Engineering — Completed
- Computer Vision 101 — Applied AI | CNNs, Image Filters & Object Detection — Completed
- GAN for Cardiovascular Disease (CVD) — Infosys Springboard | Medical Deep Learning — Completed
- Introduction to Deep Learning — Deep Learning Track | Neural Network Architectures & Backpropagation — Completed
- Introduction to NLP (Natural Language Processing) — Applied AI | Tokenization, Embeddings & Transformers — Completed
- AWS Educate — Machine Learning Foundations — AWS Educate | Feature Pipelines & Model Evaluation — Completed
- AWS Academy Graduate — Cloud Foundations — AWS Academy | Cloud Architecture, Model Serving & Infrastructure — Completed
- Programming in Python with AI — Technical Education | Object-Oriented Design & Microservices — Completed
- Introduction to Robotic Process Automation — Automation Track | Software Bots & Workflow Automation — Completed
- Prime 2.0: AI/ML Batch — Advanced Training | ML, Deep Learning, GenAI, RAG Systems, PyTorch — In Progress

Achievements:

- Winner, College Project Expo (1st Place): Awarded 1st place for designing, developing, and demonstrating an IoT-based Smart Irrigation & Rain Alert System, showcasing its sensory pipeline and agricultural water conservation impact.
- Official Patent Lead: Filed official Indian Patent Application No. 202641109078 for an AI-based multimodal crop health detection, forecasting, and adaptive agronomic management system.
- Secretary — Robotics & Automation Lab (RAL Club): Directing club activities, technical bootcamps, drone demonstrations, and embedded automation showcases for 200+ engineering students at MLRITM.
- Smart India Hackathon (SIH): Active participant and team contributor designing real-time AI automated solutions for complex challenges.

Ideas & Currently Exploring:

- AI Plant Health Platform: Computer Vision + Agentic AI for plant leaf disease detection, continuous monitoring, and automated agronomy decision support (Deep Learning, Zero-Shot Learning, Agriculture Vision).
- Ultron — AI Desktop Assistant: Voice-controlled AI agents for desktop automation, converting speech into autonomous OS controls and multi-step computer workflows (Speech AI, PyAutoGUI, Tool Calling).
- Agentic AI Workflows: Researching and experimenting with autonomous AI agents, multi-tool calling, structured reasoning loops, and automated task execution (LangGraph, Tool Calling, RAG).
- Currently Learning & Mastering: Agentic AI, LangGraph, Advanced Computer Vision, Transfer Learning, FastAPI Deployment, and Voice AI.